What Can Machine Learning Predict in NASDAQ Markets?

Price, Direction and Volatility Forecasting Across Models and Timeframes

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AI, Machine Learning and Big Data for Finance and Management · Waseda University
Final report · November 2026 · E-mini Nasdaq-100 futures (NQ), 2006–2025

Executive Summary

Volatility is clearly predictable; for returns and candle direction no useful improvement was demonstrated

Problem Markets are hard to forecast, but are all their characteristics equally hard? For the next NQ futures candle we ask what machine learning can predict: the return, the direction or the realized volatility.

Volatility

PREDICTABLE

+40% / +52%

less error than "next = current" (5 min / 1 hour candles)

Also beats HAR-OLS (+6.8% / +26.1%); rise or fall called right 72% / 75% (majority 51% / 48%)

Direction

MARGINAL

+0.69 pp

accuracy over always-the-majority-class: 50.59% vs 49.90% (DNN 256-128-64-32, 5 min)

About 7 extra right calls per 1,000 candles; seen in 2021–2023, gone in 2024–2025

Return

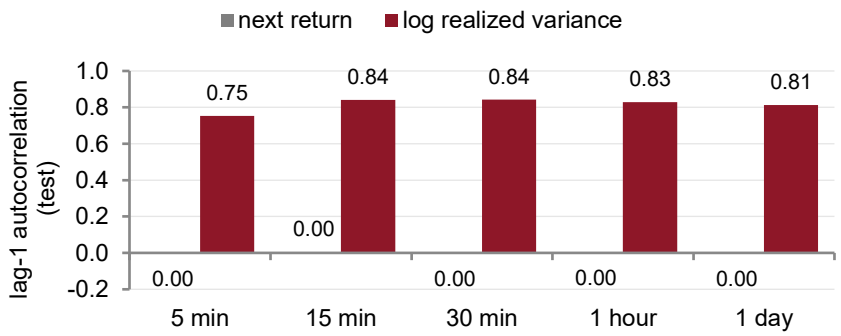
NOT DEMONSTRATED

R^2 −0.04%

vs simply forecasting zero, 95% interval [−0.10%, +0.02%] (DNN 256-128-64-32, 5 min)

Forecasts move by less than ±0.5 bp while a typical 5 min return is about 7.7 bp: the signal is lost in the noise

Why: volatility clusters, returns do not



Method

NQ futures 2006–2025, 5 candle lengths (5 min–1 day); 15 models, from benchmarks to deep networks and Mistral's Timestral. Train 2006–17, select 2017–21, test 2021–25.

Innovation

One pipeline, one data set and one test period for three different targets, so "what is predictable" is answered with the same models on every task and timeframe.

Usefulness

Volatility forecasts feed risk limits, position sizing and volatility-sensitive decisions. Direction and return forecasts at these horizons are too weak to act on.

Conclusion

Realized volatility is the most predictable characteristic; for returns and direction no useful improvement was demonstrated with the models, data and horizons tried.

Exploratory: the test period (2021-07 → 2025-05; daily to 2025-01) was examined in earlier project phases and the volatility selection score was revised after a first look at test results. Intervals: 95% bootstrap in 20-session blocks. pp = percentage points, bp = 0.01%.

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Data & Methodology

One instrument, five candle lengths, three targets, one chronological split

NASDAQ historical data
NQ 1-min, 2006–2025

→ **Feature engineering**
lags, momentum, volatility, time of day

→ **Models**
benchmarks · ML · DNN · Timestral

→ **Forecasts**
for candle t+1

→ **Out-of-sample evaluation**
test from 2021-07 (exploratory)

Three targets, all for the next candle (t+1), decided at the close of t

Target	Definition	Evaluated with
1 Next return	r = log(C(t+1)/C(t)), regression	MAE, RMSE, out-of-sample R^2 vs zero forecast
2 Next-candle direction	UP if C(t+1) > O(t+1); doji kept, counts as a miss	accuracy, balanced accuracy, F1 vs majority class
3 Future realized volatility	RV = Σ (1-min log returns)² inside candle t+1	log-RV error (R^2) vs persistence; RV UP if RV(t+1) > RV(t)

- **Instrument:** E-mini Nasdaq-100 futures (NQ, CME), back-adjusted continuous series. No QQQ, MNQ, cash index or VIX.
- **Split:** train 2006-01-10 → 2017-08-24, validation → 2021-07-14, test 2021-07-15 → 2025-05-30 (daily test ends 2025-01-30).
- **Leakage control:** features use candles ≤ t only; scaling fitted on training data; labels never cross a session or a split boundary.
- **Selection:** hyper-parameters and the five configurations chosen on validation; volatility score revised after a first look at test numbers, so not blind: results are exploratory.

Data profile per timeframe (test sample)

Candle	Rows: train · val · test	UP / DOWN / doji	Return s.d. (bp)	ACF return	ACF log RV
5 min	769k · 269k · 273k	50% / 48% / 1.8%	7.7	−0.005	**+0.75**
15 min	264k · 88k · 90k	51% / 48% / 1.0%	13.3	+0.003	**+0.84**
30 min	133k · 45k · 44k	51% / 48% / 0.8%	18.5	−0.002	**+0.84**
1 hour	65k · 22k · 22k	52% / 48% / 0.5%	26.2	−0.000	**+0.83**
1 day	1,991 · 822 · 642	54% / 46% / 0.0%	116.9	−0.001	**+0.81**

Reading: candle direction is close to a coin flip (UP 50%–54%); ACF (lag-1 autocorrelation; intraday pairs within a session) of returns is about zero (−0.005 to +0.003), while log realized variance is strongly persistent (0.75 to 0.84). Features (23): past returns, momentum, rolling volatility, range, body, wicks, RSI, moving-average gaps, relative volume, time of day, past log RV (1, 5, 22 candles, same slot on 5 prior sessions). Timestral sees the last 512 closes.

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All Models Tested

15 models on 5 timeframes: 150 model–timeframe–task configurations, all run on the same data

Family	Model	Method	Tasks
Baselines	**Zero return / training mean**	naive forecast	R
	Majority class	always the most frequent training class	D
	Persistence	RV(t+1) = RV(t)	V
	Rolling mean (22 candles)	average past variance	V
Statistical	**HAR-OLS**	linear regression on past log RV (1, 5, 22 candles, same time slot)	V
	Ridge regression	L2-penalised linear, α chosen on validation	R · V
	Logistic regression	L2-penalised, C chosen on validation	D
Machine learning	**Random forest**	200 trees, depth 8	R · D · V
	XGBoost	gradient boosting, depth 4, early stopping	R · D · V
Deep learning	**DNN 64-32-16 (original)**	feed-forward, ReLU, AdamW, early stopping	R · D · V
	DNN 64-32		R · D · V
	DNN 128-64-32		R · D · V
	DNN 256-128-64-32		R · D · V
Mistral AI	**Timestral-5M, frozen weights**	zero-shot quantiles of the next close from 512 past closes; median → return; direction = median vs last close (target: next candle's body)	R · D
	Timestral-5M + supervised calibration	q10–q90 spread → log RV via a linear map fitted on validation targets (on validation: fitted on earlier sessions only)	V

Notes

- **R** return · **D** direction · **V** volatility
- **Mistral's role:** Timestral is a time-series foundation model (local weights, no API, no training here). It is not a language model and it does not read news. Scored on 20,000 random validation and 20,000 test origins per timeframe (all daily ones).
- **Timestral outcome:** frozen weights beat no benchmark on any timeframe (return, direction); the calibrated volatility variant beat persistence only at 5 min. Neither is among the five; per-timeframe numbers in the report tables.
- **Considered, not run:** LSTM (the feature networks reach their early-stopping optimum within a few epochs). Timestral 28M / 123M / 209M: weights available locally, not tested in this study.
- **Not used:** Mistral chat / LLM API — no such integration exists in the repository.

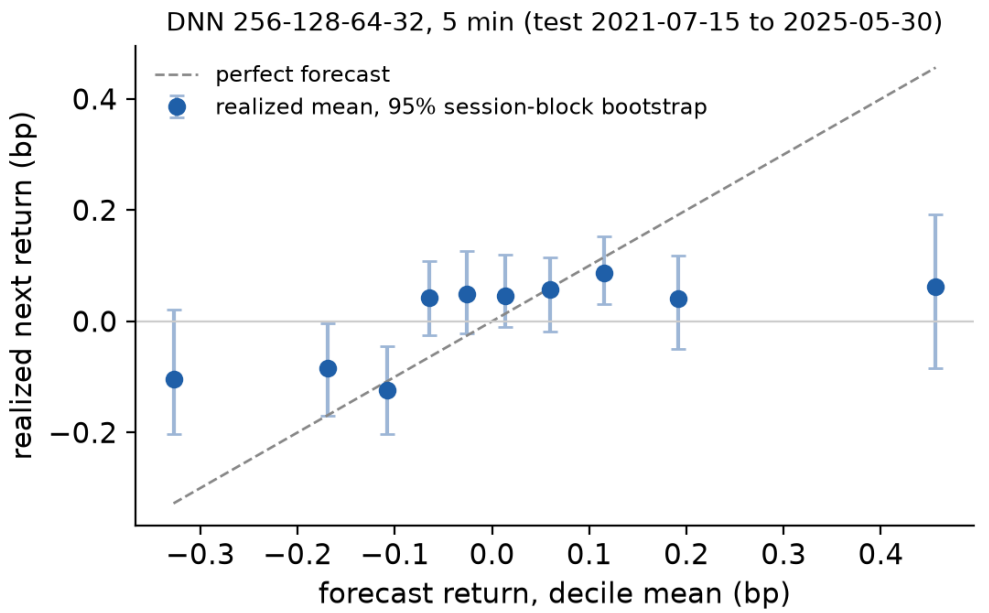
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Price / Return Forecasting

No improvement over the zero-return forecast demonstrated with the models, data and horizons tried

#	Model, timeframe	MAE (bp)	RMSE (bp)	RMSE zero (bp)	R^2 vs zero	95% interval	Sign right
1	**DNN 256-128-64-32, 5 min**	4.49	7.71	7.71	**−0.043%**	[−0.103%, +0.016%]	51.3%

- **Result:** DNN 256-128-64-32, 5 min: R^2 −0.043%, interval not distinguishable from zero. A positive R^2 would mean a smaller squared error than always forecasting zero.
- **Predicted vs realized (right):** realized returns per forecast decile. The forecasts span less than ±0.5 bp, while a typical candle moves about 7.7 bp (RMSE): the signal is buried in noise.
- **Economic reading:** no usable return signal was found with the models, data and horizons tried. This is consistent with near-random-walk returns at these horizons, but does not prove it. All return models and Timestral were run on all five timeframes (inventory, slide 4).



bp = basis point = 0.01%. R^2 interval: model and zero-forecast errors resampled together in 20-session blocks, ratio recomputed per replicate. Decile bars: 20-session block bootstrap within each decile. Test 2021-07-15 → 2025-05-30 (exploratory: examined in earlier project phases).

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Next-Candle Direction Classification

A small gain over the majority class (+0.69 pp, interval above zero) that is not stable across test years

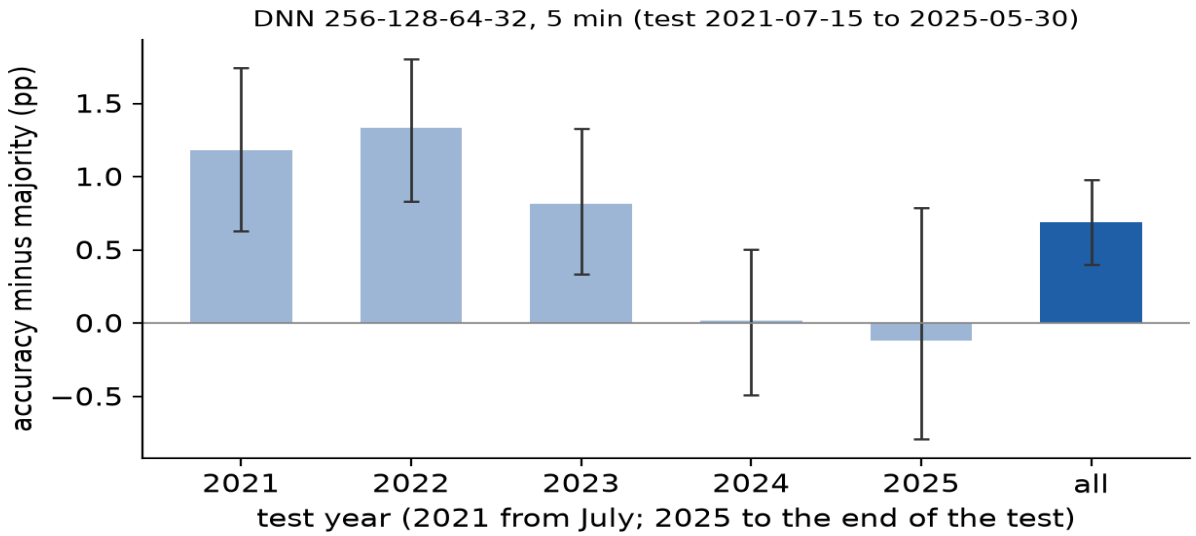
#	Model, timeframe	Accuracy	Balanced acc.	Majority	Gain	95% interval	Precision UP	Recall UP	F1 UP
2	**DNN 256-128-64-32, 5 min**	**50.59%**	51.37%	49.90%	**+0.69 pp**	[+0.40, +0.98]	52.0%	60.4%	0.559
3	**DNN 64-32, 15 min**	**50.99%**	50.83%	50.74%	**+0.25 pp**	[−0.07, +0.57]	51.8%	78.0%	0.622

Confusion matrix, DNN 256-128-64-32, 5 min (test)

	Actual UP	Actual DOWN
Predicted UP	**82,434** **30.7%**	76,180 28.4%
Predicted DOWN	53,956 20.1%	**55,837** **20.8%**

Non-doji test candles. Dojis (open = close), counted as misses: 4,933. Predicted UP on 59.1% of candles.

- **Size:** +0.69 pp is about 7 more right calls per 1,000 candles; balanced accuracy 51.37%.



- **Stability (right):** gain above zero in 2021, 2022, 2023; interval includes zero in 2024, 2025. The overall gain comes from the early test years.

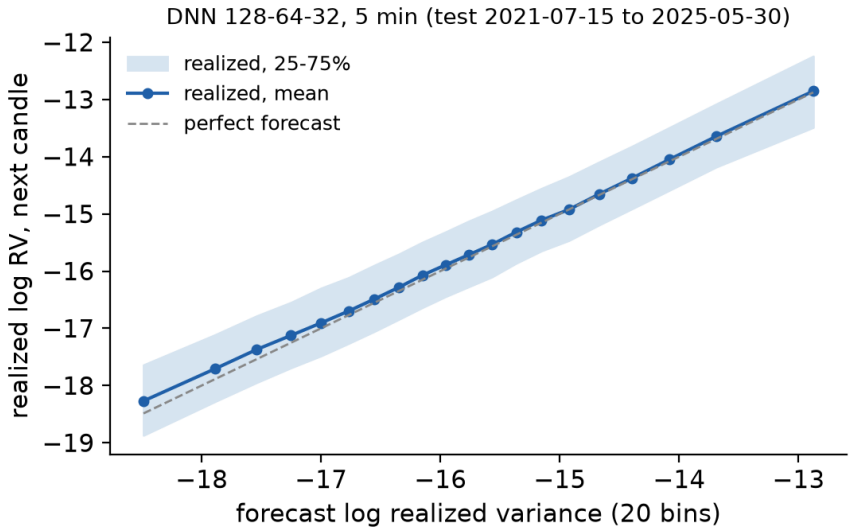
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Volatility Forecasting

Large gain over persistence, smaller but positive gain over the HAR-OLS benchmark (exploratory)

#	Model, timeframe	RMSE log RV	Persist.	HAR-OLS	R^2 vs persistence	R^2 vs HAR-OLS	RV dir. acc.	Balanced	Majority	HAR dir. acc.	Dir. gain vs HAR
4	**DNN 128-64-32, 5 min**	**0.901**	1.163	0.933	**+40.0% [+39.5, +40.4]**	**+6.8% [+6.2, +7.3]**	**72.2%**	72.0%	51.1%	71.0%	**+1.13 pp [+0.95, +1.31]**
5	**XGBoost, 1 hour**	**0.572**	0.825	0.666	**+52.0% [+50.0, +54.1]**	**+26.1% [+23.8, +28.5]**	**74.8%**	75.1%	47.7%	70.5%	**+4.28 pp [+3.53, +4.88]**

- **Method:** realized variance of the next candle from its 1-minute squared log returns (r^2 is a variance proxy; volatility = √variance). Models forecast log RV; R^2 = share of the benchmark's squared log-RV error removed. QLIKE check, model / HAR / persistence: 0.455 / 0.477 / 1.258; 0.264 / 0.275 / 0.559.
- **Two benchmarks:** persistence is naive; HAR-OLS already captures clustering, mean reversion and the time-of-day pattern (RV direction 71.0% / 70.5%). The gain vs HAR, not vs persistence, is what ML adds.
- **Options (qualitative):** rising realized volatility favours long straddles / strangles, falling favours short ones; but option prices embed IMPLIED volatility, so a correct realized-volatility forecast is not a profitable options trade. No option prices or P&L were computed.



Selection score: initially QLIKE vs persistence; revised to log-RV squared error after first test numbers were printed (QLIKE was dominated by a few limit-locked March 2020 candles): not blind. R^2 intervals: model and benchmark errors resampled together in 20-session blocks; direction gains: paired block bootstrap.

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Which NASDAQ Market Characteristic Is Most Predictable?

Five configurations chosen with validation data (volatility score revised after a first look at test results), scored on the exploratory test

ID	Task	Model	Timeframe	Key metric (test)	Baseline	Improvement [95% interval]
1	**Return**	DNN 256-128-64-32	5 min	RMSE 7.71 bp	Zero return: RMSE 7.71 bp	**R^2 -0.04% [-0.10, +0.02]**
2	**Direction**	DNN 256-128-64-32	5 min	Accuracy 50.59%	Majority: 49.90%	**+0.69 pp [+0.40, +0.98]**
3	**Direction**	DNN 64-32	15 min	Accuracy 50.99%	Majority: 50.74%	**+0.25 pp [-0.07, +0.57]**
4	**Volatility**	DNN 128-64-32	5 min	log-RV RMSE 0.901; RV UP/DOWN 72.2%	Persistence 1.163; majority 51.1%; HAR 0.933, 71.0%	**vs persistence R^2 +40.0% [+39.5, +40.4]; vs HAR R^2 +6.8% [+6.2, +7.3], RV dir. +1.13 pp**
5	**Volatility**	XGBoost	1 hour	log-RV RMSE 0.572; RV UP/DOWN 74.8%	Persistence 0.825; majority 47.7%; HAR 0.666, 70.5%	**vs persistence R^2 +52.0% [+50.0, +54.1]; vs HAR R^2 +26.1% [+23.8, +28.5], RV dir. +4.28 pp**

Strongest return approach
DNN 256-128-64-32, 5 min: R^2 −0.043%, not distinguishable from zero: no improvement over zero demonstrated with the models, data and horizons tried.

Strongest direction approach
DNN 256-128-64-32, 5 min: +0.69 pp over the majority class, interval above zero; not distinguishable from zero in 2024, 2025.

Strongest volatility approach
log-RV R^2 vs persistence / vs HAR-OLS: DNN 128-64-32, 5 min +40% / +6.8%; XGBoost, 1 hour +52% / +26.1%. RV direction above HAR by +1.13 pp / +4.28 pp.

Exploratory verdict: realized volatility is the most predictable characteristic. Its gain is large over persistence and smaller but positive over HAR-OLS, with intervals above zero on both selected timeframes, consistent with the strong persistence of log realized variance in the data. For returns no improvement over zero was demonstrated; the direction gain is under one percentage point and not stable over time. Metrics differ by task, so the ranking rests on benchmark-relative gains and their uncertainty.

Originality one pipeline for three targets, five timeframes, classical, deep and foundation models. **Usefulness** volatility forecasts for risk limits and sizing. **Feasibility** runs on a laptop CPU with free local models; reproducible (next slide).

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Reproducibility

All tables, charts and slides are regenerated from the code in the repository; the licensed NQ data are not published

Code, results and this deck https://github.com/Paciooo/Project_ai_ML_bigdata · branch main

Step	What	Command (PowerShell, repository root)
1 Environment	Python 3.12, Node.js 18+	`pip install -r requirements.txt -r requirements-timestral.txt`
2 Data (not in Git)	NQ 1-minute bars 2006–2025 (TradeStation, licensed); Timestral-5M weights from the local Hugging Face cache	`$env:FUTURES_DATA_DIR = "C:\path\to\futures-data"`
3 Train and forecast	all models, five timeframes, CPU only	`python multitarget.py --tag final`
4 Tables and charts	results_*.csv, data_profile.csv, selection, charts	`python multitarget_report.py <run> reports\multitarget_final`
5 Independent check	recomputes every number from the saved forecasts	`python verify_multitarget.py <run> reports\multitarget_final presentation`
6 Tests and deck	unit tests; slides from deck_data.json	`python -m pytest -q` · `cd presentation; npm ci; node build_deck.js`

What has been checked

- verify_multitarget.py: every metric, the selection rule, the data profile and the deck data recomputed independently; 0 problems on the final run
- unit tests for targets, bootstrap, selection rule and the causal Timestral calibration
- run 20261112T050812Z_82953a_final

Limits

- the test period had been examined before: results are exploratory
- without the licensed NQ data the training cannot be re-run; the saved forecasts are not published either
- the run was made from a working copy; run.json stores the hash of every source file

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